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Aim of the Experiment

Implementation of a comprehensive image captioning system using custom CNN-LSTM architectures.

1 Introduction

Image captioning represents one of the most challenging intersection problems in artificial intelligence, requiring systems to automatically generate natural language descriptions that accurately represent visual content. This task demands deep understanding of both visual semantics and linguistic structures, bridging the gap between computer vision and natural language processing.

1.1 What is Image Captioning?

Image captioning generates textual descriptions summarizing image content. Unlike classification (single label) or object detection (identifying objects), captioning requires understanding relationships between objects, their attributes, and scene context to produce coherent sentences.

Mathematically, given an input image I, the goal is to generate a word sequence $S = \{w_1, w_2, ..., w_n\}$ maximizing:

$$P(S|I) = \prod_{t=1}^{n} P(w_t|w_1, w_2, ..., w_{t-1}, I) \tag{1}$$

This transforms captioning into sequence generation where each word depends on both image features and previous words.

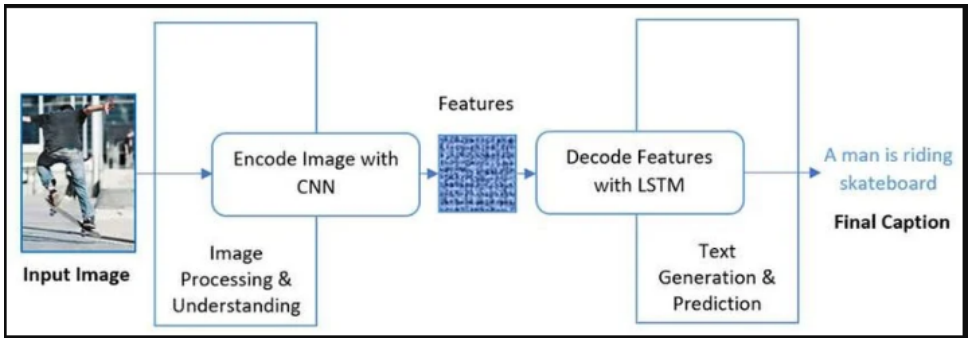


Figure 1: Example of Image Captioning: Given an image, the model generates descriptive text[13]

1.2 Applications and Importance

Image captioning bridges the semantic gap between visual perception and human understanding, enabling critical applications:

- **Accessibility:** Assists visually impaired individuals by converting visual information to spoken descriptions, enabling independent navigation and environmental understanding

- **Content Retrieval:** Powers image search engines and digital asset management through natural language queries, significantly improving search accuracy
- **Healthcare:** Assists radiologists by generating preliminary reports from medical imaging (X-rays, CT scans, MRI), highlighting abnormalities and reducing diagnostic time

1.3 Challenges in Image Captioning

Image captioning faces fundamental bottlenecks from both visual understanding and language generation:

1.3.1 Visual Understanding Challenges

- **Object Recognition Complexity:** Images contain multiple objects with varying scales, orientations, and partial occlusions, requiring robust feature extraction in cluttered scenes
- **Scene Context and Relationships:** Models must comprehend spatial relationships and interactions; distinguishing "a person riding a horse" versus "standing next to a horse" requires relational reasoning
- **Visual Ambiguity:** Fine-grained discrimination between similar objects, understanding object states, and recognizing subtle attributes (colors, textures, emotions) pose significant challenges
- **Out-of-Vocabulary Concepts:** Models trained on limited datasets may fail on rare objects or novel visual concepts, producing generic captions

1.3.2 Natural Language Processing Challenges

- **Semantic Gap:** The fundamental challenge lies in bridging continuous visual features (high-dimensional Euclidean spaces) with discrete linguistic representations (symbolic, rule-governed domains)
- **Linguistic Coherence:** Captions must be grammatically correct, semantically meaningful, and stylistically natural, requiring learning of syntactic structures and discourse conventions
- **Variable Complexity:** Different images require descriptions of varying length and detail; models must dynamically determine appropriate elaboration levels
- **Caption Diversity:** Single images can be described in multiple valid ways; models often produce generic descriptions rather than capturing unique image aspects

1.4 Evolution of Image Captioning Approaches

Image captioning has evolved through distinct phases, each introducing novel architectures:

1.4.1 Traditional Statistical Approaches (Pre-2015)

- **Template-Based:** Predefined sentence templates filled with detected objects and attributes (e.g., "___ is ___ near ___")
- **Retrieval-Based:** Query images matched with visually similar database images; captions transferred or modified based on detected differences
- **Limitations:** Produced rigid, unnatural captions limited to predefined patterns without generalization to novel compositions

1.4.2 Neural Encoder-Decoder Architectures (2015-2018)

- **CNN-RNN Framework:** CNNs encode visual features; RNNs (particularly LSTMs) decode into caption sequences word by word conditioned on encoded features
- **Attention Mechanisms:** "Show, Attend and Tell" (Xu et al., 2015) introduced dynamic spatial attention, enabling fine-grained alignment between words and image regions
- **Integration Strategies:**
  - Inject: Image features initialize RNN hidden state (our approach)
  - Merge: Image features concatenated with word embeddings at each time step

1.4.3 Attention-Enhanced Approaches (2018-present)

- **Bottom-Up Attention:** Anderson et al. (2018) used object detection features with task-specific attention, grounding captions on detected objects rather than uniform grid features
- **Adaptive Attention:** Models learning when to attend visual features versus linguistic context, improving handling of non-visual and content words
- **Semantic Attention:** Explicit prediction of visual attributes and relationships as intermediate steps before caption generation

2 System Requirements

This section outlines hardware specifications, software dependencies, and environment configurations necessary for implementation.

2.1 Dataset Requirements

The project utilizes the Flickr8k dataset and Google News Word2Vec embeddings:

- **Flickr8k Dataset:** 8,091 images with 5 captions each (40,455 total captions)
- **Image Format:** JPG, various resolutions (resized to 224×224)
- **Storage:** Approximately 1.1 GB for images
- **Word2Vec:** GoogleNews-vectors-negative300.bin.gz (1.5 GB compressed, 3.6 GB uncompressed)
- **Vocabulary:** 3 million words/phrases with 300-dimensional vectors

2.2 Hardware Requirements

Table 1 summarizes minimum and recommended hardware specifications.

Table 1: Hardware Requirements

Component	**Minimum Requirements**	**Recommended Requirements**
Processor (CPU)	Intel Core i5 (8th Gen) / AMD Ryzen 5	Intel Core i7/i9 / AMD Ryzen 7/9
RAM	8 GB DDR4	16-32 GB DDR4
Graphics (GPU)	NVIDIA GTX 1050 Ti (4 GB VRAM)	NVIDIA RTX 3060/4060+ (8+ GB VRAM)
Storage	20 GB available space	50 GB SSD
Internet	Required for dataset/model downloads	High-speed broadband

Note on GPU Requirements: While technically runnable on CPU-only systems, GPU acceleration through CUDA-enabled NVIDIA graphics cards is strongly recommended. Training on CPU can be 50-100x slower, making the 20-epoch training schedule impractical.

2.3 Software Requirements

Table 2: Software and Library Requirements

Category	Library/Framework	Version
Core ML Framework	PyTorch	2.0+
	torchvision	0.15+
	CUDA Toolkit	11.7+ (for GPU)
	cuDNN	8.5+ (for GPU)
Deep Learning	torch.nn	Built-in
	torch.optim	Built-in
	torch.utils.data	Built-in
	torchvision.models	Built-in
NLP Libraries	NLTK	3.8+
	Gensim	4.3+
	Word2Vec (Google News)	Pretrained vectors
Data Processing	NumPy	1.24+
	Pandas	2.0+
	Pillow (PIL)	9.5+
	Regular Expressions (re)	Built-in
Visualization	Matplotlib	3.7+
	Tabulate	0.9+
	torchviz	0.0.2
SOTA Baseline	Transformers (HuggingFace)	4.30+
	BLIP Model (Salesforce)	Pretrained
Environment	Python	3.9-3.11
	Jupyter/Google Colab	Latest

2.4 Development Environment

The project was developed using Google Colaboratory (Colab), a cloud-based Jupyter environment providing free GPU access:

- **GPU Accessibility:** Free access to NVIDIA Tesla T4, P100, or K80 GPUs
- **Pre-installed Libraries:** PyTorch, NumPy, Pandas come pre-installed
- **Reproducibility:** Notebooks easily shared and executed identically
- **Storage Integration:** Google Drive integration for persistent storage

3 Flowchart Diagram

This section presents comprehensive data flow diagrams for the image captioning system organized into three phases: Data Preparation, Model Training, and Evaluation.

3.1 Phase 1: Data Creation and Preprocessing

The data creation phase transforms raw Flickr8k data into a standardized augmented dataset. Figure 2 illustrates this pipeline.

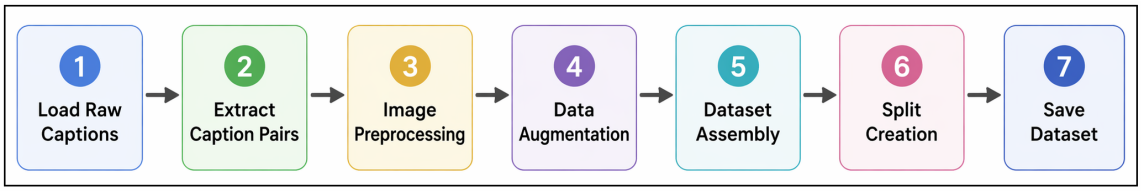


Figure 2: Data Creation Flowchart: Raw inputs through preprocessing to processed dataset

Pipeline Steps:

1. **Load Raw Captions:** Read captions.txt (40,455 entries, 5 captions per image)
2. **Extract Caption Pairs:** For each image, select shortest and longest caption by token count, creating pairs $\{(I_i, S_i^{short}, S_i^{long})\}_{i=1}^{8091}$
3. **Image Preprocessing:** Load from /images directory, convert to RGB, resize to 224×224, convert to PyTorch tensor
4. **Data Augmentation:** Generate 2 versions per image (original and horizontal flip), expanding dataset from 8,091 to 16,182 samples
5. **Dataset Assembly:** Stack augmented image tensors, store caption pairs and token counts
6. **Split Creation:** Random permutation (seed=42), 80% training, 10% validation, 10% testing
7. **Save Dataset:** Save as flickr8k_dataset_raw.pt containing images, captions, splits, and metadata

3.2 Phase 2: Model Training

The second phase constructs and trains the hybrid CNN-LSTM architecture on captions.

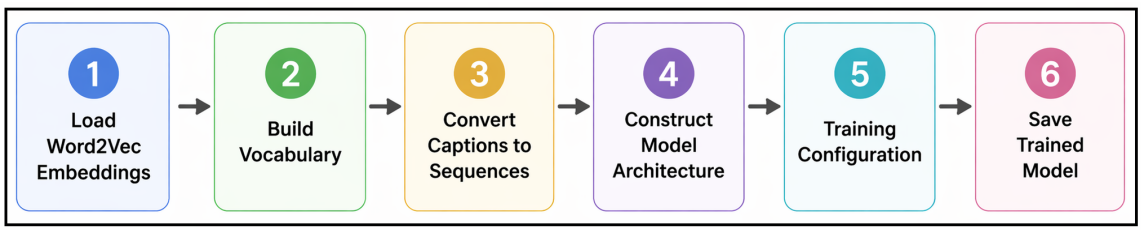


Figure 3: Model Training Flowchart: Dataset loading through model saving

Training Process:

1. **Load Word2Vec:** Google News embeddings (300-dim) for semantic word representation
2. **Build Vocabulary:** Tokenize captions, create word-to-index mapping with special tokens (`<pad>`, `<start>`, `<end>`, `<unk>`)
3. **Convert Captions:** Map tokens to indices, add markers, pad/truncate to max_len=20
4. **Construct Architecture:**
   - Encoder: Frozen ResNet18 (pretrained on ImageNet), output 512-dim features
   - Embedding: 300-dim layer initialized with Word2Vec (frozen)
   - Decoder: 2-layer LSTM (hidden=512, dropout=0.3)
   - Output: Linear layers (512→512→vocab_size)
5. **Training Configuration:** CrossEntropyLoss, Adam (lr=0.001), ReduceLROnPlateau scheduler, teacher forcing ratio=0.5, batch size=64, 20 epochs
6. **Save Model:** Save checkpoint as image_captioning_model.pt

3.3 Phase 3: Evaluation

The third phase performs comparative evaluation against the baseline BLIP model.

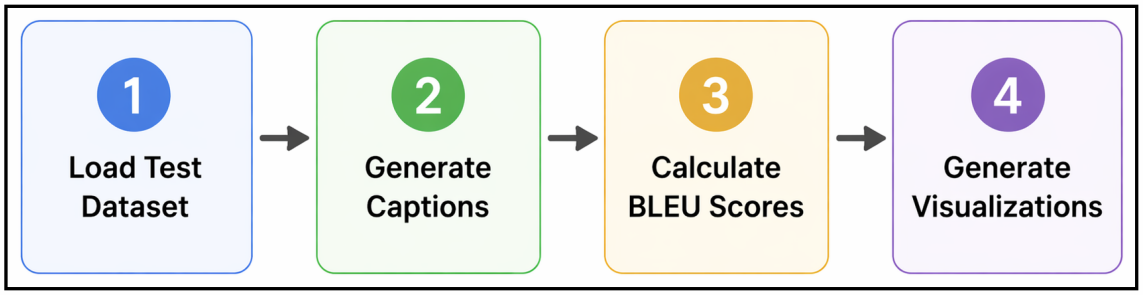


Figure 4: Evaluation Flowchart: Comparative analysis of custom models against BLIP baseline

Evaluation Process:

1. Load test dataset (10% split from Phase 1)
2. Generate captions from custom models and BLIP
3. Calculate BLEU-1 scores on 50 test images against ground truth
4. Generate visualizations: side-by-side comparisons, BLEU bar charts, architecture summaries

4 Code

This section presents key implementations from the image captioning system's four primary scripts.

4.1 Code Structure Overview

Table 3: Code File Organization

File	Purpose	Key Functions/Classes
data_creation.ipynb	Data preprocessing	preprocess_caption(), count_tokens(), apply_augmentations()
hybrid.ipynb	Caption training	ImageCaptioningModel, CaptionIndexDataset
hybrid_long.ipynb	Long caption training	Same architecture with extended parameters
visuals.ipynb	Evaluation & comparison	BLEU computation, visualization generation

4.2 Data Preprocessing Module

The preprocessing pipeline handles caption cleaning, tokenization, and augmentation:

```
def preprocess_caption(caption) -> List[str]:
    """Clean and tokenize a caption"""
    caption = caption.lower()
    caption = re.sub(r'[^a-z0-9\s\']', '', caption)
    tokens = nltk.tokenize.word_tokenize(caption)
    tokens = [token for token in tokens
              if token.strip() and token not in ["'", '"', "''", "``"]]
    return tokens

def count_tokens(text):
    """Count tokens in preprocessed text"""
    return len(nltk.word_tokenize(text.lower()))
```

Listing 1: Caption preprocessing and tokenization

The caption selection process extracts shortest and longest captions per image to enable controlled complexity experiments:

```
for image_name in captions_df['image'].unique():
    image_captions = captions_df[captions_df['image'] == image_name]['
   caption'].tolist()
    shortest = min(image_captions, key=count_tokens)
    longest = max(image_captions, key=count_tokens)
    image_data.append({
        'image': image_name,
        'short_caption': shortest,
        'long_caption': longest
    })
```

Listing 2: Selecting caption pairs for each image

4.3 Data Augmentation Module

Augmentation expands the dataset through geometric transformations. Only horizontal flipping is applied:

```python
def apply_augmentations(image_tensor):
    """Generate 2 augmented versions of an image:
    1. Original
    2. Horizontal flip
    """
    augmented_images = []
    augmented_images.append(image_tensor)  # Original
    augmented_images.append(transforms.functional.hflip(image_tensor))
    return augmented_images

# Base transformations applied to all images
base_transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
])
```

Listing 3: Image augmentation implementation

4.4 Model Architecture

The hybrid CNN-LSTM architecture combines pretrained components with custom decoders:

```python
class ImageCaptioningModel(nn.Module):
    def __init__(self, vocab_size, embedding_dim=300, hidden_dim=512):
        super().__init__()

        # Image encoder (frozen ResNet18 pretrained on ImageNet)
        self.encoder = models.resnet18(pretrained=True)
        for param in self.encoder.parameters():
            param.requires_grad = False  # Freeze encoder weights
        in_features = self.encoder.fc.in_features
        self.encoder.fc = nn.Identity()  # Remove classification head
        self.image_projection = nn.Linear(in_features, hidden_dim)

        # Word embedding with semantic initialization
        self.word_embedding = nn.Embedding(vocab_size, embedding_dim)
        self.init_embeddings_with_word2vec()

        # LSTM decoder with regularization
        self.lstm = nn.LSTM(embedding_dim, hidden_dim,
                            batch_first=True, num_layers=2, dropout=0.3)

        # Output prediction layers
        self.output_layer = nn.Sequential(
            nn.Linear(hidden_dim, 512),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(512, vocab_size)
        )
```

Listing 4: Complete model architecture

4.5 Word2Vec Initialization

Pretrained embeddings provide semantic knowledge transfer:

```
def init_embeddings_with_word2vec(self):
    """Initialize embedding layer with pretrained Word2Vec vectors"""
    print("Initializing embeddings with Word2Vec...")
    initialized = 0
    for word, idx in word_to_idx.items():
        if word in w2v_model:
            self.word_embedding.weight.data[idx] = torch.tensor(
    w2v_model[word])
            initialized += 1
    print(f"Initialized {initialized}/{len(word_to_idx)} words with
    Word2Vec")
    self.word_embedding.weight.requires_grad = False  # Freeze
    embeddings
```

Listing 5: Word2Vec embedding initialization

4.6 Training Loop

The training process implements teacher forcing with dynamic scheduling:

```
# Loss and optimization
criterion = nn.CrossEntropyLoss(ignore_index=word_to_idx['<pad>'])
optimizer = optim.Adam(model.parameters(), lr=0.001)
scheduler = optim.lr_scheduler.ReduceLROnPlateau(
    optimizer, patience=3, factor=0.5)

num_epochs = 20
teacher_forcing_ratio = 0.5

# Training loop structure
for epoch in range(num_epochs):
    model.train()
    for batch in train_loader:
        images = batch['image'].to(device)
        target_indices = batch['caption_indices'].to(device)

        outputs = model(images, target_indices, teacher_forcing_ratio
    =0.5)
        loss = criterion(outputs.permute(0, 2, 1), target_indices)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

    scheduler.step(val_loss_avg)
```

Listing 6: Training configuration and loop

5 Output and Results

This section presents comprehensive output from the image captioning system, including model summaries, training progress, qualitative examples, and quantitative evaluation.

5.1 Model Architecture Summaries

Metric	Value
Total Parameters	18,569,300
Trainable Parameters	7,392,788
Non-trainable Parameters	11,176,512
Model Size (FP32)	70.84 MB

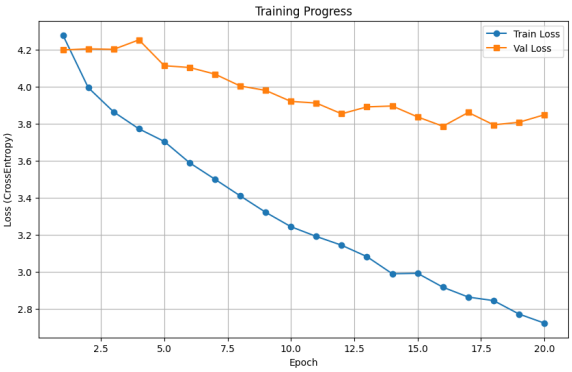
(a) Model Summary

Layer Name	Layer Type	Parameters
encoder	ResNet	11,176,512
image_projection	Linear	262,656
word_embedding	Embedding	1,143,600
lstm	LSTM	3,768,320
output_layer	Sequential	2,218,212

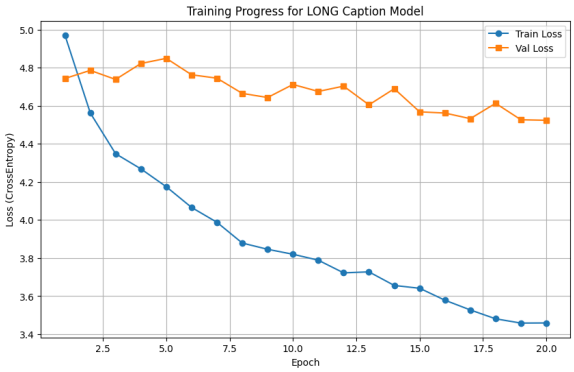
(b) Model Architecture

Figure 5: Model architecture summaries showing parameter counts and configurations

5.2 Training Progress



(a) Caption Model Training



(b) Long Caption Model Training

Figure 6: Training and validation loss curves over 20 epochs

5.3 Qualitative Results



Ground Truth (Short): A light-colored dog runs on the beach .

Custom Short Model: a dog running in the snow

Ground Truth (Long): A white dog is running down a rocky beach .

Custom Long Model: a dog is running through the water with a stick in its mouth

BLIP (SOTA): a dog running on the beach with a ball in its mouth



Ground Truth (Short): A tennis player hitting a ball .

Custom Short Model: a man in a basketball

Ground Truth (Long): The tennis player is hitting the ball with his racquet , while others watch the match .

Custom Long Model: a man in a white shirt is a a a a a a a

BLIP (SOTA): a man playing tennis



Ground Truth (Short): A motorcycle daredevil making a high jump .

Custom Short Model: a man on a

Ground Truth (Long): a lone motorcyclist jumping in midair with his legs in the air and sparks flying in the background

Custom Long Model: a man is a a a a a a a a

BLIP (SOTA): a man on a bike doing a trick

Figure 7: Caption comparison: Ground truth vs. generated from custom models and BLIP

5.4 BLEU Score Evaluation

Table 4: Short Caption BLEU Score Comparison

Metric	**Our Model**	**Custom**	**BLIP (SOTA)**
BLEU-1 Score	[0.2309]		[0.3345]
Improvement	-		[+10.37]%

Table 5: Long Caption BLEU Score Comparison

Metric	**Our Model**	**Custom**	**BLIP (SOTA)**
BLEU-1 Score	[0.2151]		[0.2141]
Improvement	-		[-0.09]%

6 Contributions

Table 6: Team Member Contributions

Name	Roll Number	Contributions
Prashant	2447021	• Researched and selected the Flickr8k dataset for image captioning task • Finalized the hybrid model approach combining ResNet18, LSTM, and pretrained Word2Vec embeddings • Designed and implemented the custom architecture for long caption generation • Evaluated and compared custom model performance against BLIP using BLEU scores • Developed an interactive Gradio web interface for real-time caption generation
Nikita	2447027	• Loaded and preprocessed the Flickr8k dataset by extracting shortest captions and tokenizing text • Designed and implemented the custom CNN-LSTM architecture for short caption generation with Word2Vec embeddings • Trained the short caption model for 20 epochs using cross-entropy loss and Adam optimizer • Visualized training and validation loss curves to monitor model convergence • Documented the complete final report

7 Limitations

While the implemented hybrid CNN-LSTM architecture demonstrates fundamental image captioning principles, several limitations merit acknowledgment:

1. **Limited Visual Feature Extraction:**
    - ResNet18 extracts only global image features without spatial attention, preventing selective focus on specific regions during word generation
    - Modern architectures (Faster R-CNN, Vision Transformers) provide region-level features that significantly improve caption grounding
2. **Sequential Processing Bottleneck:**
    - LSTM decoder processes tokens sequentially, limiting parallelization during training and inference
    - Transformer architectures with self-attention process entire sequences simultaneously and better capture long-range dependencies
3. **Frozen Encoder Weights:**
    - ResNet encoder remains frozen during training, preventing task-specific fine-tuning of visual features
    - This reduces computational requirements but limits adaptation for captioning-specific visual nuances
4. **Dataset Scale Constraints:**
    - Flickr8k contains only 8,091 images compared to MS-COCO (330K) or Conceptual Captions (3.3M)
    - Limited diversity restricts generalization to novel visual compositions
5. **Evaluation Metric Limitations:**
    - BLEU-1 only measures unigram overlap, ignoring syntactic correctness and semantic adequacy
    - High word overlap but incorrect word order can produce misleadingly high scores
    - Modern metrics (CIDEr, SPICE, BERTScore) provide more semantically meaningful evaluation
6. **Computational Resource Constraints:**
    - Google Colab's free GPU tier limits batch sizes and total training epochs
    - Word2Vec embeddings (3.6 GB uncompressed) require significant memory
    - More extensive hyperparameter tuning could potentially improve performance

8 Observations

Based on implementation and evaluation, the following key observations were recorded:

8.1 Model Training Behavior

- **Convergence Patterns:** Both caption models showed steady training loss decrease over 20 epochs, with validation loss plateauing after approximately 12-15 epochs, indicating primary pattern learning
- **Word2Vec Initialization Impact:** Pretrained embeddings provided strong semantic initialization; even frozen, the model effectively mapped visual features to semantically meaningful word sequences, with related words appropriately clustered
- **Scheduler Effectiveness:** ReduceLROnPlateau effectively reduced learning rates when validation loss stagnated, preventing overfitting in later epochs

8.2 Comparative Performance Insights

- **Caption Quality:** Short caption model generally produced more grammatically correct outputs due to simpler targets; long model occasionally generated more descriptive but less coherent captions
- **Generic Caption Tendency:** Both models sometimes produced safe, generic captions (e.g., "a man is standing on a field") for diverse images, indicating learned frequency biases
- **BLIP Comparison:** BLIP consistently outperformed both custom models, demonstrating transformer effectiveness and large-scale pretraining benefits; performance gap larger for long captions where attention mechanisms better captured complex relationships
- **Simple Scene Performance:** For prominent objects in simple scenes, custom models occasionally produced results comparable to BLIP

8.3 Generation Characteristics

- **Error Patterns:** Common errors included object misidentification, missing important scene elements, and hallucination (describing objects not present), particularly in ambiguous scenes
- **Exposure Bias Effects:** Teacher forcing during training may have contributed to inference difficulties where models struggled with their own predictions
- **Augmentation Benefits:** Horizontal flipping effectively increased training data diversity, helping prevent overfitting and improving generalization to varied orientations

9 Conclusion

This project successfully implemented and evaluated a hybrid CNN-LSTM architecture for automatic image captioning, demonstrating fundamental principles of bridging computer vision and natural language processing.

9.1 Key Achievements

1. **End-to-End Pipeline Implementation:**
   - Developed complete pipeline from raw data preprocessing through model training to comprehensive evaluation
   - Implemented data augmentation strategies enhancing the limited Flickr8k dataset
   - Successfully integrated pretrained components (ResNet18, Word2Vec) with custom LSTM decoders
2. **Dual Caption Generation Capability:**
   - Trained separate models for short and long caption generation
   - Demonstrated trade-offs between caption brevity and descriptive richness
   - Provided flexible captioning options for different application contexts
3. **Comprehensive Evaluation:**
   - Conducted both qualitative (caption examples) and quantitative (BLEU scores) evaluation
   - Established baseline comparisons with state-of-the-art BLIP model
   - Identified specific strengths and weaknesses through systematic error analysis

9.2 Technical Insights

The project reinforced several important principles in modern deep learning:

- Transfer learning effectiveness through pretrained encoders and embeddings significantly reduces data requirements
- Data preprocessing and augmentation play critical roles in small-dataset scenarios
- Sequence generation faces inherent challenges including exposure bias and evaluation metric limitations
- A significant performance gap exists between classical encoder-decoder architectures and modern transformer-based approaches
- The integration of vision and language modalities requires careful architectural design choices

In conclusion, this project demonstrates that while modern transformer architectures have set new benchmarks in image captioning, classical CNN-LSTM approaches remain valuable for understanding fundamental concepts and serve as practical baselines for educational and resource-constrained applications.

10 Learning Outcomes

This project provided comprehensive exposure to multiple aspects of deep learning, computer vision, and natural language processing:

- **Deep Learning Framework Proficiency:** Hands-on PyTorch experience for building, training, and evaluating neural networks; understanding automatic differentiation, gradient computation, and optimization algorithms; implementation of custom Dataset and DataLoader classes
- **Computer Vision Concepts:** Understanding CNN architectures (ResNet) and feature extraction; application of transfer learning through pretrained models; implementation of data augmentation techniques for improved generalization
- **Natural Language Processing:** Working with word embeddings (Word2Vec) and distributed representations; sequence modeling with LSTM networks for text generation; vocabulary building, tokenization, and text preprocessing
- **Multimodal Learning:** Integrating visual and textual modalities in unified neural architectures; understanding challenges of cross-modal alignment and translation; comparing inject vs. merge approaches for multimodal fusion
- **Experimental Design:** Designing controlled experiments with short vs. long caption models; proper train/validation/test splitting strategies; setting up comparative baselines (BLIP) for meaningful evaluation
- **Metric Understanding:** Critical analysis of BLEU scores and their limitations; understanding differences between quantitative metrics and human-perceived quality; interpreting loss curves to diagnose training issues
- **Resource Management:** Efficient GPU memory utilization through appropriate batch sizing; managing large pretrained models and datasets within memory constraints; implementing model checkpointing for training resumption
- **Debugging Skills:** Diagnosing shape mismatches in tensor operations; understanding gradient flow and vanishing gradient issues; debugging vocabulary indexing and padding/packing issues
- **Cloud Computing:** Utilizing Google Colab for GPU-accelerated training; managing cloud storage for datasets and model checkpoints; understanding trade-offs between local and cloud-based development
- **Fundamental AI Concepts:** Encoder-decoder paradigm for sequence generation; teacher forcing and its impact on training stability vs. inference; role of pretraining in modern deep learning workflows
- **Limitations Awareness:** Understanding where classical architectures fall short compared to SOTA; recognizing data, computational, and algorithmic constraints; critical thinking about evaluation methodology

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